UNIT 6 · STANDARD 6.AT.7

Generate Equivalent Expressions

Lesson 6-8

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.

Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Properties of Operations · **Commutative Property** · **Associative Property** · **Identity Property** · **Property** · **Distributive Property** · **Equivalent expressions** · **Expand** · **Factor an expression**

Level 3: try the blanks from memory first, then check the bank.

1 Rules that let you reorder or regroup without changing the _____.

2 The rule that order does not change a sum or product, like a + b = b + a, is the _____.

3. The rule that grouping does not change a sum or product, like (a + b) + c = a + (b + c), is the _____ .

4. The rule that adding 0 or multiplying by 1 keeps a number the same is the _____ .

5. A rule that is always true in math is a _____ .

6. The rule that lets you multiply a number across a sum, like a(b + c) = a × b + a × c, is the _____ .

7. Expressions that always have the same value are _____ .

8. Writing 6(3x + 4) as 18x + 24 is called _____ the expression.

9. Writing 18x + 24 as 6(3x + 4) is called _____ .

Write About the Math

What properties did the engineer use, and why does rearranging make the calculation easier?
¿Qué propiedades usó la ingeniera, y por qué reacomodar hace más fácil el cálculo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Properties of Operations** *Propiedades de las operaciones*
- ☐ **Commutative Property** *Propiedad conmutativa*
- ☐ **Associative Property** *Propiedad asociativa*
- ☐ **Identity Property** *Propiedad de identidad*
- ☐ **Property** *Propiedad*

Model: *The commutative property lets you change the order of a sum without changing the total.* — Students name the commutative property and explain that 3 + 5 + 2 gives 10 in any order because reordering addends does not change the sum.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The engineer used the ___ property to change the ___ and the ___ property to change the ___. This makes it easier because 4 × 25 = ___, which is a friendly number to multiply by.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.